

Notes: Nevo (Econometrica, 2001)

Measuring Market Power in the Ready-to-Eat Cereal Industry

Contents

1	Big picture	3
2	Incremental contribution of this paper	3
3	Data and measurement	4
3.1	Industry background	4
3.2	Scanner data and markets	4
3.3	Product characteristics and demographics	4
3.4	Observed margins	5
4	Models and empirical specifications	5
4.1	Random-coefficients indirect utility	5
4.2	Taste heterogeneity	5
4.3	Market shares	5
4.4	Logit benchmark	6
4.5	Demand estimation and instruments	6
4.6	Multiproduct Nash-Bertrand pricing	6
4.7	Supply conduct comparison	6
5	Identification and empirical design	7
5.1	What identifies demand	7
5.2	Why brand fixed effects matter	7
5.3	What identifies conduct	7
5.4	Main limits	7
6	Tables and figures	8
6.1	Table I: Volume Market Shares and Table II: Aggregate Estimates of Production Costs	8
6.2	Table III: Detailed Estimates of Production Costs and Table IV: Prices and Market Shares of Brands in Sample	8
6.3	Table V: Results from Logit Demand	9
6.4	Table VI: Results from the Full Model	9
6.5	Figure 1: Frequency Distribution of Taste for Sogginess and Figure 2: Frequency Distribution of Price Coefficient	10
6.6	Table VII: Median Own and Cross-Price Elasticities	11
6.7	Table VIII: Median Margins	12
7	Short summary	12

8	Conclusion	12
8.1	Core conclusion	12
8.2	Economic intuition	12
8.3	Incremental contribution revisited	13
8.4	What the paper changes for empirical IO	13
8.5	Limitations and open questions	13
8.6	Takeaway	13

1 Big picture

- **Question.** Are the high price-cost margins in the U.S. ready-to-eat cereal industry evidence of collusive pricing, or can they be explained by product differentiation and multiproduct pricing by firms with brand portfolios?
- **Core empirical object.** The paper estimates brand-level demand for 25 major cereal brands across city-quarter markets and uses the estimated own- and cross-price elasticities to infer markups under alternative supply conduct models.
- **Main conclusion.** High observed margins are not enough to infer collusion. In this industry, a multi-product Nash-Bertrand model with differentiated products is broadly consistent with observed accounting margins.
- **Main decomposition.** Observed margins are separated into three forces: product differentiation, portfolio pricing by multiproduct firms, and potential price collusion.
- **Demand model.** The paper estimates a random-coefficients discrete-choice model, rather than relying on simple Logit substitution patterns.
- **Data advantage.** The data are a panel of prices and quantities for leading cereal brands across up to 65 cities and 20 quarters, which allows city-level price variation and brand fixed effects to be used in the estimation.
- **Identification advantage.** Price endogeneity is addressed using instruments based on prices of the same brand in other cities and observed cost proxies, relying on the idea that demand shocks are independent across cities once brand effects, demographics, and time effects are controlled for.
- **Key demand result.** The full random-coefficients model produces intuitive substitution patterns: kid cereals substitute more strongly toward other kid cereals, and nutritional cereals substitute more strongly toward similar nutritional cereals.
- **Key supply result.** Current ownership of the 25 major brands yields a median margin around 42.2% in the full model, while joint ownership of those brands would imply a much higher margin around 72.6%.
- **Policy implication.** Large accounting margins should not mechanically be interpreted as collusion. In differentiated-products industries, high margins may arise from brand loyalty, advertising-created perceived quality, and multiproduct pricing.

2 Incremental contribution of this paper

- **From concentration to conduct.** Earlier work often viewed the cereal industry as a classic example of cooperative pricing because it was concentrated, profitable, advertising-intensive, and full of brand proliferation. This paper moves the discussion from descriptive structure to structural conduct measurement.
- **Margin decomposition.** The paper does not simply estimate markups. It decomposes the observed high markups into product differentiation, portfolio pricing, and residual collusive conduct.
- **Demand-side flexibility.** It brings the Berry-Levinsohn-Pakes/random-coefficients discrete-choice framework to the cereal industry and shows that restrictive Logit substitution can lead to misleading conclusions about market power.
- **Panel-data identification with brand fixed effects.** The paper uses city-quarter variation and brand fixed effects to control for unobserved product quality, rather than relying entirely on product-characteristic exogeneity.
- **Demographic heterogeneity.** It incorporates the empirical distribution of demographics rather than imposing only parametric taste heterogeneity, allowing price sensitivity and taste for characteristics to differ systematically across households.

- **Supply-side testing rather than assumption.** Many structural IO exercises assume a conduct model. This paper estimates demand first and then evaluates which supply model best matches observed margins.
- **Antitrust relevance.** The paper provides a template for evaluating whether high margins reflect anti-competitive conduct or equilibrium pricing with differentiated products.

3 Data and measurement

3.1 Industry background

- The ready-to-eat cereal industry is highly concentrated. The top three firms have around 70–75% volume share in the sample period, while the top six firms account for close to 90% or more.
- The industry has high advertising-to-sales ratios and frequent product introductions.
- Prior antitrust concerns focused on both pricing and brand proliferation.
- The central empirical challenge is to determine whether high price-cost margins reflect collusion or non-collusive pricing of differentiated brands.

Interpretation: The industry is a useful test case because it combines high concentration, many differentiated products, and rich product-level data.

3.2 Scanner data and markets

- Prices and market shares are from the IRI Infoscan data, aggregated by brand, city, and quarter.
- A market is a city-quarter.
- The sample covers the first quarter of 1988 through the last quarter of 1992.
- The main demand estimates use the 25 brands with the highest national market shares in the last quarter of 1992.
- For almost all brands there are 1,124 market observations.
- The 25 brands account for around 43–62% of city-level cereal volume and 55–60% of national cereal volume.

Interpretation: The panel dimension is central: price variation across cities and time is used to identify demand while controlling for brand effects.

3.3 Product characteristics and demographics

- Brand characteristics include price, advertising, calories from fat, sugar, fiber, whether a cereal is mushy, and segment dummies such as kids, adults, and all-family.
- Demographics come from the Current Population Survey and are matched to city-quarter markets.
- The model uses income, income squared, age, household size, and child indicators to generate heterogeneity in preferences.

Interpretation: Demographics matter because the model lets price sensitivity and taste for characteristics vary across consumers. This is what permits substitution patterns to depend on economic similarity rather than just market shares.

3.4 Observed margins

- The paper uses accounting and cost estimates to form crude measures of observed price-cost margins.
- Table II gives aggregate cost information for the cereal industry and the food sector.
- Table III gives more detailed estimates of production costs, marketing expenses, and operating profits.
- The retail margin estimate is around 46%, while the manufacturer price-cost margin is high relative to many food industries.

Interpretation: The observed margins are imperfect but useful benchmarks. The supply models are evaluated by asking which predicted margins are close to these observed margins.

4 Models and empirical specifications

4.1 Random-coefficients indirect utility

Consumer i 's indirect utility from cereal brand j in market t is written as

$$u_{ijt} = \alpha_i p_{jt} + x'_{jt} \beta_i + \xi_j + \Delta \xi_{jt} + \varepsilon_{ijt}. \quad (1)$$

Here p_{jt} is price, x_{jt} are observed product characteristics, ξ_j is a brand fixed effect, $\Delta \xi_{jt}$ is a market-specific unobserved demand shock, and ε_{ijt} is an idiosyncratic taste shock.

Interpretation: The brand fixed effect captures persistent unobserved quality, such as brand image or reputation. The market-specific shock is what creates price endogeneity.

4.2 Taste heterogeneity

The taste parameters vary across consumers:

$$\begin{pmatrix} \alpha_i \\ \beta_i \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \Pi D_i + \Sigma v_i, \quad (2)$$

where D_i are demographics and v_i are random taste shocks.

Interpretation: This is the key difference from simple Logit. Consumers are allowed to differ in price sensitivity and in their valuation of cereal characteristics.

4.3 Market shares

Let A_{jt} be the set of consumer characteristics for which product j is the utility-maximizing option. The model-implied market share is

$$s_{jt}(x_t, p_t, \delta_t; \theta_2) = \int_{A_{jt}} dP^*(D, v, \varepsilon). \quad (3)$$

Interpretation: Market shares are integrals over heterogeneous consumers. They are computed numerically because the full random-coefficients model does not have closed-form Logit shares.

4.4 Logit benchmark

The simple Logit model implies substitution patterns that depend primarily on market shares. If two products have similar shares, Logit treats them as similar substitutes even if their characteristics are very different.

Interpretation: This is a crucial weakness in the cereal setting. A kid cereal should be a closer substitute for another kid cereal than for a plain nutritional cereal, even if the two candidate substitutes have similar market shares.

4.5 Demand estimation and instruments

The demand error is the unobserved market-specific product quality shock $\Delta\xi_{jt}$. Prices are endogenous because firms observe some component of this shock when setting prices. The GMM estimation uses instruments satisfying

$$E[Z'_{jt}\Delta\xi_{jt}] = 0. \quad (4)$$

The key instruments are:

- prices of the same brand in other cities within the same region;
- observed cost proxies;
- brand and time dummy variables in the relevant specifications.

Interpretation: Regional cost shocks make prices correlated across cities, while city-specific demand shocks are assumed independent across cities after controls. This makes other-city prices useful instruments.

4.6 Multiproduct Nash-Bertrand pricing

For a multiproduct firm f , the first-order condition for price p_j is

$$s_j(p) + \sum_{k \in \mathcal{F}_f} (p_k - mc_k) \frac{\partial s_k(p)}{\partial p_j} = 0. \quad (5)$$

In matrix form, for a given ownership structure,

$$p - mc = \Delta^{-1}s, \quad (6)$$

where Δ contains demand derivatives for products owned by the same firm.

Interpretation: A multiproduct firm internalizes substitution among its own brands. This portfolio effect raises margins even without collusion.

4.7 Supply conduct comparison

The paper computes implied margins under several conduct models:

- single-product firms;
- current ownership of the 25 brands;
- joint ownership of the 25 brands;
- current ownership of all brands;
- monopoly/perfect price collusion.

Interpretation: The model that produces predicted margins closest to observed accounting margins is interpreted as the most plausible conduct benchmark.

5 Identification and empirical design

5.1 What identifies demand

- Within-brand variation in price across cities and quarters identifies price sensitivity.
- Brand fixed effects absorb persistent brand quality.
- Demographic variation across markets helps identify how tastes vary with consumer characteristics.
- Other-city prices and cost proxies provide instruments for local prices.

Interpretation: The identifying assumption is that a city-specific demand shock for a brand is not correlated with the same brand's prices in other cities, once common brand and time components are controlled for.

5.2 Why brand fixed effects matter

- Cereal brands differ in unobserved ways that are likely important for demand.
- If unobserved brand quality is omitted, price coefficients may be biased because high-quality brands can charge high prices.
- Brand fixed effects allow the demand estimation to use within-brand price variation.

Interpretation: This is an important incremental contribution relative to earlier differentiated-products demand estimation: product fixed effects reduce reliance on observed product characteristics to explain all quality differences.

5.3 What identifies conduct

- Demand estimates imply elasticities.
- Elasticities imply markups under different ownership/conduct structures.
- The paper compares these predicted markups with observed accounting margins.

Interpretation: Conduct is not identified by a conduct parameter estimated directly inside demand. It is inferred by comparing the margin implications of alternative supply models.

5.4 Main limits

- Observed margins are crude accounting benchmarks.
- The paper rules out the extreme joint-profit-maximization model, but does not rule out every possible form of partial collusion.
- The model treats product characteristics, advertising, and product introduction as given.
- The paper does not directly model brand proliferation or advertising as strategic entry deterrence.

Interpretation: The paper is strongest on price conduct conditional on the existing product set. It is not a complete model of dynamic product positioning or entry deterrence.

6 Tables and figures

6.1 Table I: Volume Market Shares and Table II: Aggregate Estimates of Production Costs

Read:

- Table I shows that the top three cereal firms held roughly 72–75% of market volume during 1988–1992.
- The top six firms held roughly 88–95% of volume, confirming that the industry is highly concentrated.
- Private label share grew from 3.33% in 1988Q1 to 7.60% in 1992Q4, but remained small relative to the major firms.
- Table II shows that RTE cereal has a gross margin of 64.4% using aggregate production-cost data, compared with 26.5% for the broader food sector.

Economic message: High concentration and high accounting margins motivate the collusion question, but they do not by themselves identify collusion.

TABLE I
VOLUME MARKET SHARES

	88Q1	88Q4	89Q4	90Q4	91Q4	92Q4
Kellogg	41.39	39.91	38.49	37.86	37.48	33.70
General Mills	22.04	22.30	23.60	23.82	25.33	26.83
Post	11.80	10.30	9.45	10.96	11.37	11.31
Quaker Oats	9.93	9.00	8.29	7.66	7.00	7.40
Ralston	4.86	6.37	7.65	6.60	5.45	5.18
Nabisco	5.32	6.01	4.46	3.75	2.95	3.11
C3	75.23	72.51	71.54	72.64	74.18	71.84
C6	95.34	93.89	91.94	90.65	89.58	87.53
Private Label	3.33	3.75	4.63	6.29	7.13	7.60

Source: IRI Infoscan Data Base, University of Connecticut, Food Marketing Center.

Table I: Volume Market Shares

TABLE II
AGGREGATE ESTIMATES OF PRODUCTION COSTS

Item	RTE Cereal (SIC 2043)		All Food Industries (SIC 20)	
	M\$	% of value	M\$	% of value
Value of Shipments	8,211	100.0	371,246	100.0
Materials	2,179	26.5	235,306	63.4
Labor	677	8.2	32,840	8.8
Energy	76	0.9	4,882	1.3
Gross Margin		64.4		26.5

Source: Annual Survey of Manufacturers 1988–1992.

Table II: Aggregate Estimates of Production Costs

6.2 Table III: Detailed Estimates of Production Costs and Table IV: Prices and Market Shares of Brands in Sample

Read:

- Table III decomposes manufacturer price into production costs, marketing expenses, and operating profits.
- The estimated retail-price gross margin is 46.0%, which becomes an important benchmark for supply-model predictions.
- Table IV reports a mean price of 19.4 cents per serving and a mean brand share within the cereal market of 2.2%.
- Most price variation is across brands, but city variation in prices remains important for identification.

Economic message: Table III supplies the observed-margin benchmark. Table IV shows why the city-quarter panel has enough variation to estimate demand.

TABLE III
DETAILED ESTIMATES OF PRODUCTION COSTS

Item	\$/lb	% of Mfr Price	% of Retail Price
Manufacturer Price	2.40	100.0	80.0
Manufacturing Cost:	1.02	42.5	34.0
Grain	0.16	6.7	5.3
Other Ingredients	0.20	8.3	6.7
Packaging	0.28	11.7	9.3
Labor	0.15	6.3	5.0
Manufacturing Costs (net of capital costs) ^a	0.23	9.6	7.6
Gross Margin		57.5	46.0
Marketing Expenses:	0.90	37.5	30.0
Advertising	0.31	13.0	10.3
Consumer Promo (mfr coupons)	0.35	14.5	11.7
Trade Promo (retail in-store)	0.24	10.0	8.0
Operating Profits	0.48	20.0	16.0

^a Capital costs were computed from ASM data.

Source: Cotterill (1996) reporting from estimates in CS First Boston Reports "Kellogg Company," New York, October 25, 1994.

Table III: Detailed Estimates of Production Costs

TABLE IV
PRICES AND MARKET SHARES OF BRANDS IN SAMPLE

Description	Mean	Median	Std	Min	Max	Brand Variation	City Variation	Quarter Variation
Prices (¢ per serving)	19.4	18.9	4.8	7.6	40.9	88.4%	5.3%	1.6%
Advertising (M\$ per quarter)	3.56	3.04	2.03	0	9.95	66.2%	—	1.8%
Share within Cereal Market (%)	2.2	1.6	1.6	0.1	11.6	82.3%	0.5%	0%

Source: IRI Infoscan Data Base, University of Connecticut, Food Marketing Center.

Table IV: Prices and Market Shares of Brands in Sample

6.3 Table V: Results from Logit Demand

Read:

- Table V reports OLS and IV Logit demand estimates.
- OLS price coefficients are much smaller in magnitude than IV estimates, consistent with price endogeneity from unobserved demand shocks.
- The preferred Logit IV specifications produce large negative price coefficients, roughly between -22 and -24 in the relevant columns.
- First-stage statistics show that the instruments are highly predictive of prices in the reported specifications.
- The Logit model is used as a benchmark, but its substitution restrictions are too strong for the conduct exercise.

Economic message: The Logit estimates illustrate the importance of IV, but the model's substitution patterns can distort inferred markups because cross-price elasticities depend mechanically on shares.

6.4 Table VI: Results from the Full Model

Read:

- Table VI estimates the random-coefficients model with brand fixed effects, observed product characteristics, and demographic interactions.
- The mean price coefficient is large and negative, around -27.2 .
- Demographic interactions are important: older and higher-income households are less price sensitive.
- The estimated share of consumers with positive price coefficients is only 0.7%, indicating that the demographic specification avoids economically implausible price responses for almost all consumers.
- The minimum-distance fit of brand effects to observed characteristics is rejected, implying that brand fixed effects capture important unobserved product quality.

Economic message: The full model provides the elasticities needed for the supply-side exercise. It lets substitution depend on both product characteristics and consumer demographics.

Variable	OLS			IV					
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)
Price	-4.96 (0.10)	-7.26 (0.16)	-7.97 (0.15)	-8.17 (0.11)	-17.57 (0.50)	-17.12 (0.49)	-22.56 (0.51)	-23.77 (0.53)	-23.07 (0.47)
Advertising	0.158 (0.002)	0.026 (0.002)	0.026 (0.002)	0.157 (0.002)	0.020 (0.002)	0.020 (0.002)	0.018 (0.002)	0.017 (0.002)	0.018 (0.002)
Log of Median Income	—	—	0.89 (0.02)	—	—	—	1.06 (0.02)	1.13 (0.02)	1.12 (0.02)
Log of Median Age	—	—	-0.423 (0.052)	—	—	—	-0.063 (0.059)	0.003 (0.062)	-0.007 (0.061)
Median HH Size	—	—	-0.126 (0.027)	—	—	—	-0.053 (0.029)	-0.036 (0.031)	-0.038 (0.031)
Fit/Test of Over Identification ^b	0.54	0.72	0.74	436.9 (26.30)	168.5 (30.14)	181.2 (16.92)	83.96 (30.14)	82.95 (16.92)	85.87 (42.56)
1st Stage R ²	—	—	—	0.889	0.908	0.908	0.910	0.909	0.913
1st Stage F-test	—	—	—	5119	124	288	129	144	180
Instruments ^c	—	—	—	brand dummies	prices	cost	prices	prices, cost	prices, cost

^aDependent variable is $\ln(S_{i,t}) - \ln(S_{i,t-1})$. Based on 27,862 observations. All regressions include time dummy variables, and with the exception of columns (i) and (ix), all regressions also include brand dummy variables. The regressions in columns (i) and (ix) include product characteristics (calories from fat, sugar, fiber, mushy and segment dummy variables); see text for reported coefficients. The regression in column (ix) includes city dummy variables. Asymptotically robust *s.e.* are reported in parentheses.

^bAdjusted R² for the OLS regressions, and a test of over identification for the IV regressions (Hausman (1983)) with the 0.95 critical values in parentheses.

^cPrices denote the average regional price of the brand; cost denotes cost prices; both are described in the text.

Table V: Results from Logit Demand

TABLE VI
RESULTS FROM THE FULL MODEL^a

Variable	Means (β 's)	Standard Deviations (σ 's)	Interactions with Demographic Variables:			
			Income	Income Sq	Age	Child
Price	-27.198 (5.248)	2.453 (2.978)	315.894 (110.385)	-18.200 (5.914)	—	7.634 (2.238)
Advertising	0.020 (0.005)	—	—	—	—	—
Constant	-3.592 ^b (0.138)	0.330 (0.609)	5.482 (1.504)	—	0.204 (0.341)	—
Cal from Fat	1.146 ^b (0.128)	1.624 (2.809)	—	—	—	—
Sugar	5.742 ^b (0.581)	1.661 (5.866)	-24.931 (9.167)	—	5.105 (3.418)	—
Mushy	-0.565 ^b (0.052)	0.244 (0.623)	1.265 (0.737)	—	0.809 (0.385)	—
Fiber	1.627 ^b (0.263)	0.195 (3.541)	—	—	—	-0.110 (0.0513)
All-family	0.781 ^b (0.075)	0.1330 (1.365)	—	—	—	—
Kids	1.021 ^b (0.168)	2.031 (0.448)	—	—	—	—
Adults	1.972 ^b (0.186)	0.247 (1.636)	—	—	—	—
GMM Objective (degrees of freedom)			5.05 (8)			
MD χ^2			3472.3			
% of Price Coefficients > 0			0.7			

^a Based on 27,862 observations. Except where noted, parameters are GMM estimates. All regressions include brand and time dummy variables. Asymptotically robust standard errors are given in parentheses.

^b Estimates from a minimum-distance procedure.

Table VI: Results from the Full Model

6.5 Figure 1: Frequency Distribution of Taste for Sogginess and Figure 2: Frequency Distribution of Price Coefficient

Read:

- Figure 1 shows the estimated distribution of taste for sogginess. Most consumers dislike soggy cereal, but a small group appears to prefer it.
- Figure 2 shows the distribution of price sensitivity. Most consumers have negative price coefficients, and the distribution is shaped by the empirical distribution of demographics.
- Together, the figures make the random-coefficients model concrete: heterogeneity is not merely a table of coefficients but an implied distribution of preferences.

Economic message: Consumer heterogeneity is essential for realistic substitution. It is the reason a price change for a kids cereal leads to stronger substitution toward other kids cereals rather than toward unrelated products.

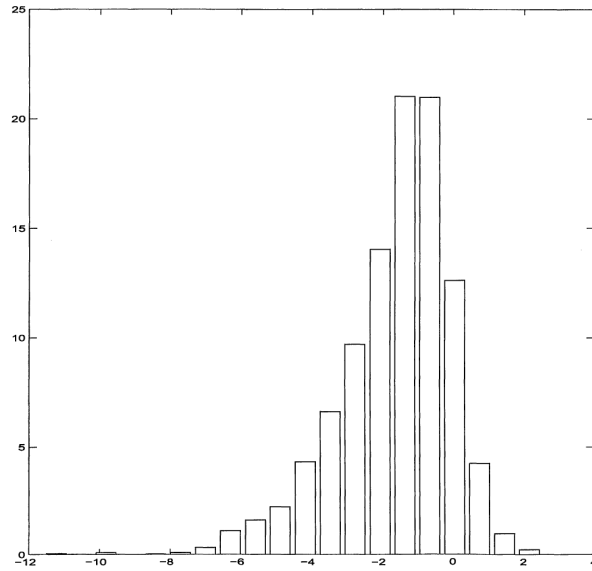


FIGURE 1.—Frequency distribution of taste for sogginess (based on Table VI).

Figure 1: Taste for Sogginess

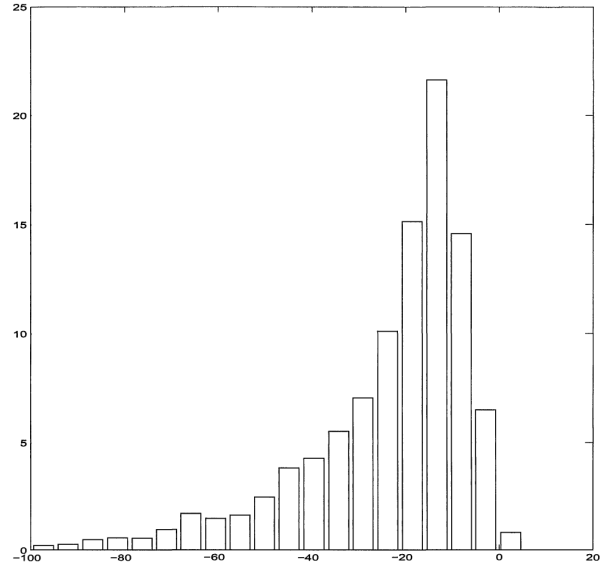


FIGURE 2.—Frequency distribution of price coefficient (based on Table VI).

Figure 2: Price Coefficient

6.6 Table VII: Median Own and Cross-Price Elasticities

Read:

- Table VII reports a sample of own- and cross-price elasticities from the full model.
- Own-price elasticities are large and negative, for example Corn Flakes around -3.38 , Frosted Flakes around -3.14 , Cheerios around -3.66 , Lucky Charms around -2.54 , and Shredded Wheat around -4.25 .
- Cross-price elasticities are not equal within columns, unlike the Logit model would imply.
- Substitution patterns are intuitive: cereals in similar segments are closer substitutes.

Economic message: Table VII is the empirical core of the paper. The margin decomposition depends on realistic own- and cross-price elasticities.

TABLE VII
MEDIAN OWN AND CROSS-PRICE ELASTICITIES^a

#	Brand	Corn Flakes	Frosted Flakes	Rice Krispies	Frost Loops	Cheerios	Total	Lucky Charms	P Raisin Bran	Cap'n Crunch	Shredded Wheat
1	K Corn Flakes	-3.379	0.212	0.197	0.014	0.202	0.097	0.012	0.013	0.038	0.028
2	K Raisin Bran	0.036	0.046	0.079	0.043	0.145	0.043	0.037	0.057	0.050	0.040
3	K Frosted Flakes	0.151	-3.137	0.105	0.069	0.129	0.079	0.061	0.013	0.138	0.023
4	K Rice Krispies	0.195	0.144	-3.231	0.031	0.241	0.087	0.026	0.031	0.055	0.046
5	K Frosted Mini Wheats	0.014	0.024	0.052	0.043	0.105	0.028	0.038	0.054	0.045	0.033
6	K Frost Loops	0.019	0.131	0.042	-2.340	0.072	0.025	0.107	0.027	0.149	0.020
7	K Special K	0.114	0.124	0.105	0.021	0.153	0.151	0.019	0.021	0.035	0.035
8	K Crispix	0.077	0.086	0.114	0.034	0.181	0.085	0.030	0.037	0.048	0.043
9	K Corn Pops	0.013	0.109	0.034	0.113	0.058	0.025	0.098	0.024	0.127	0.016
10	GM Cheerios	0.127	0.111	0.152	0.034	-3.663	0.085	0.030	0.037	0.056	0.050
11	GM Honey Nut Cheerios	0.033	0.192	0.058	0.123	0.094	0.034	0.107	0.026	0.162	0.024
12	GM Wheaties	0.242	0.169	0.175	0.025	0.240	0.113	0.021	0.026	0.050	0.043
13	GM Total	0.096	0.108	0.087	0.018	0.131	-2.889	0.017	0.017	0.029	0.029
14	GM Lucky Charms	0.019	0.131	0.041	0.124	0.073	0.026	-2.536	0.027	0.147	0.020
15	GM Tix	0.012	0.103	0.031	0.109	0.056	0.026	0.096	0.024	0.123	0.016
16	GM Raisin Nut	0.013	0.025	0.042	0.035	0.089	0.040	0.031	0.046	0.036	0.027
17	GM Cinnamon Toast Crunch	0.026	0.164	0.049	0.119	0.089	0.035	0.102	0.026	0.151	0.022
18	GM Kix	0.050	0.279	0.070	0.101	0.106	0.056	0.088	0.030	0.149	0.025
19	P Raisin Bran	0.027	0.037	0.068	0.044	0.127	0.035	0.038	-2.496	0.049	0.036
20	P Grape Nuts	0.037	0.049	0.088	0.042	0.165	0.050	0.037	0.051	0.052	0.047
21	P Honey Bunches of Oats	0.100	0.098	0.104	0.022	0.172	0.109	0.020	0.024	0.038	0.033
22	Q 100% Natural	0.013	0.021	0.046	0.042	0.103	0.029	0.036	0.052	0.046	0.029
23	Q Life	0.077	0.328	0.091	0.114	0.137	0.046	0.096	0.023	0.182	0.029
24	Q Cap'n Crunch	0.043	0.218	0.064	0.124	0.101	0.034	0.106	0.026	-2.277	0.024
25	N Shredded Wheat	0.076	0.082	0.124	0.037	0.210	0.076	0.034	0.044	0.054	-4.252
26	Outside good	0.141	0.078	0.084	0.022	0.104	0.041	0.018	0.021	0.033	0.021

^a Cell entries i, j , where i indexes row and j column, give the percent change in market share of brand i with a one percent change in price of j . Each entry represents the median of the elasticities from the 1124 markets. The full matrix and 95% confidence intervals for the above numbers are available from <http://elsa.berkeley.edu/~nevo>.

TABLE VIII
MEDIAN MARGINS^a

	Logit (Table V column ix)	Full Model (Table VI)
Single Product Firms	33.6% (31.8%–35.6%)	35.8% (24.4%–46.4%)
Current Ownership of 25 Brands	35.8% (33.9%–38.0%)	42.2% (29.1%–55.8%)
Joint Ownership of 25 Brands	41.9% (39.7%–44.4%)	72.6% (62.2%–97.2%)
Current Ownership of All Brands	37.2% (35.2%–39.4%)	—
Monopoly/Perfect Price Collusion	54.0% (51.1%–57.3%)	—

^a Margins are defined as $(p - mc)/p$. Presented are medians of the distribution of 27,862 (brand-city-quarter) observations. 95% confidence intervals for these medians are reported in parentheses based on the asymptotic distribution of the estimated demand coefficients. For the Logit model the computation is analytical, while for the full model the computation is based on 1,500 draws from this distribution.

Table VIII: Median Margins

Table VII: Median Own and Cross-Price Elasticities

6.7 Table VIII: Median Margins

Read:

- Table VIII computes predicted margins under different ownership and conduct models.
- Under the full model, single-product firms imply a median margin of 35.8%.
- Current ownership of the 25 brands implies 42.2%, close to the observed retail benchmark of around 46%.
- Joint ownership of all 25 brands implies 72.6%, much higher than the observed margin benchmark.
- This rules against the extreme full-collusion model for the 25 brands.

Economic message: The table is the main conduct test. High observed margins are consistent with noncollusive multiproduct Bertrand pricing once product differentiation is modeled flexibly.

7 Short summary

- The paper studies whether high margins in the ready-to-eat cereal industry imply collusive pricing.
- It estimates a random-coefficients discrete-choice demand system for 25 major brands across city-quarter markets.
- The model uses brand fixed effects, demographic heterogeneity, and instruments based on other-city prices and cost proxies.
- The full model produces realistic substitution patterns across cereal segments.
- Logit demand is too restrictive because it forces substitution to be based mainly on market shares.
- The paper uses demand elasticities to compute markups under different conduct models.
- Current multiproduct ownership implies median margins close to observed accounting margins.
- Full joint ownership or monopoly would imply margins much higher than observed margins.
- Therefore, high cereal margins can be explained mostly by differentiation and portfolio pricing, not necessarily collusion.
- The paper’s methods are directly relevant for merger analysis and antitrust evaluation in differentiated-products industries.

8 Conclusion

8.1 Core conclusion

Nevo shows that high price-cost margins in a concentrated industry need not imply collusive pricing. In the ready-to-eat cereal industry, observed margins are broadly consistent with noncollusive Nash-Bertrand pricing by multiproduct firms that sell differentiated brands. The paper therefore separates the existence of market power from the source of that market power.

8.2 Economic intuition

The intuition is that differentiation gives each brand some local market power, and multiproduct ownership makes firms internalize cannibalization among their own brands. Both forces raise markups without requiring explicit or tacit collusion. If consumers strongly prefer specific brands or cereal types, and if a firm owns several close substitutes, the firm can set high prices even in a noncollusive equilibrium.

8.3 Incremental contribution revisited

The incremental contribution is methodological and substantive. Methodologically, the paper shows how to combine a flexible random-coefficients demand system, brand fixed effects, panel instruments, and supply-side counterfactuals to test conduct. Substantively, it revises a classic interpretation of the cereal industry: high margins are not enough to diagnose collusion because differentiated-products demand and multiproduct pricing can generate similar margins.

8.4 What the paper changes for empirical IO

The paper is an early and influential example of using structural demand estimates to answer a competition-policy question. It demonstrates that antitrust conclusions can be very sensitive to substitution patterns. A simple Logit model might produce misleading margin predictions because it imposes unrealistic cross-price elasticities. Flexible demand estimation is therefore not only a technical improvement but also a policy necessity.

8.5 Limitations and open questions

The paper does not fully model advertising, product introduction, or brand proliferation. These dimensions were central to historical antitrust concerns in the cereal industry. The paper also does not rule out all forms of partial collusion, such as collusion among subsets of products or firms. More detailed cost data and a dynamic model of product entry and advertising would be needed to study those questions.

8.6 Takeaway

The main takeaway is that margins must be interpreted through a model of demand. In differentiated-products industries, observed price-cost margins combine the effects of consumer tastes, substitution patterns, ownership structure, and conduct. Nevo's paper provides a template for decomposing these effects and shows that careful demand estimation can overturn a simple collusion interpretation of high margins.